Case Study 2: GMRES

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Link to Github Repository:

https://github.com/mclane2/Case-Study-2-GMRES

GMRES Algorithm Test Results

The five test matrices selected from the SuiteSparse matrix collection spanning a range of sizes were nos1 (237×237), sherman1 (1000×1000), orsirr_1 (1030×1030), west0497 (497×497), and fs_541_1 (541×541).

Of the five matrices tested, only fs_541_1 converged within the maximum iteration limit of 200 iterations. For all three versions of the GMRES algorithm implemented, it reached a residual of 2.14×10^{-11} in 15 iterations. The other matrices exhausted the maximum iteration limit before reaching the tolerance. I found that the orsirr_1 (1030×1030) and west0497 (497×497) matrices were the most ill-conditioned for this problem, with still a significantly large residual after 200 iterations.

All three implementations produced identical convergence plots on every test matrix, as is seen from Figures 1-5. The first implementation was consistently at least twice as slow as versions 2 and 3 that implement Givens rotations. This because version 1 of the algorithm implements a full least squares minimization (using numpy.linalg.lstsq) at each iteration, while the other versions only have to implement a single Givens rotation per iteration. Versions 2 and 3 had similar timings, but version 3 of teh algorithm was slightly faster.

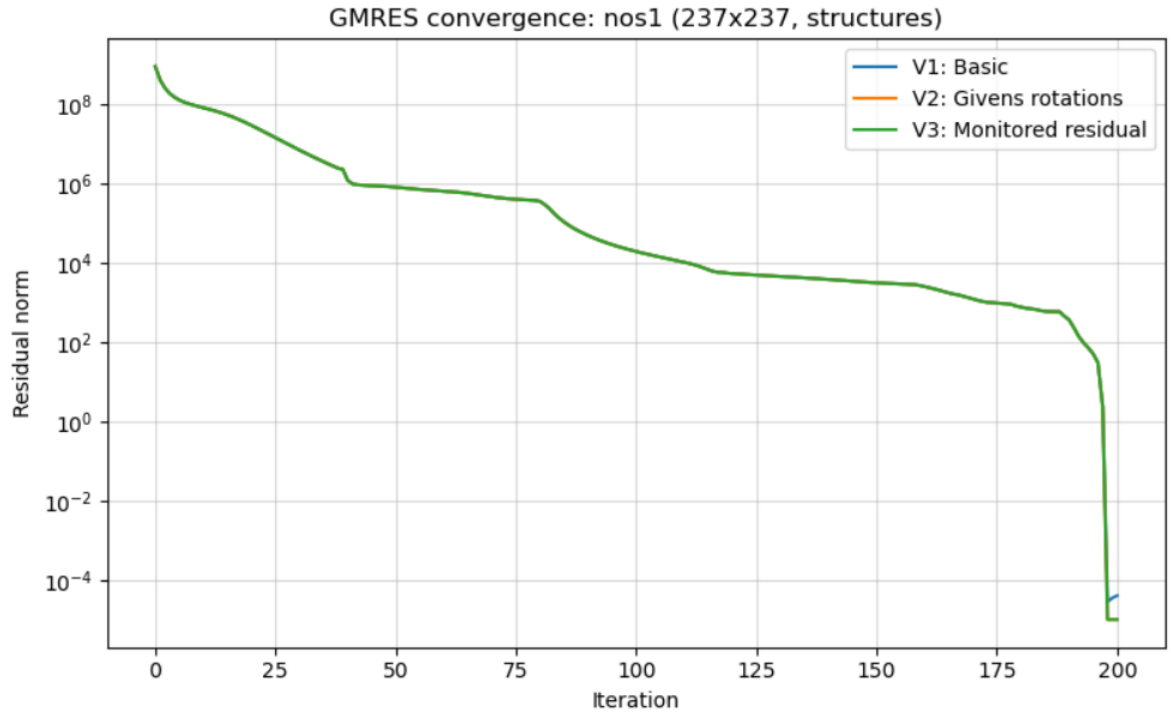

Figure 1: Convergence plot for nos1 (237×237)

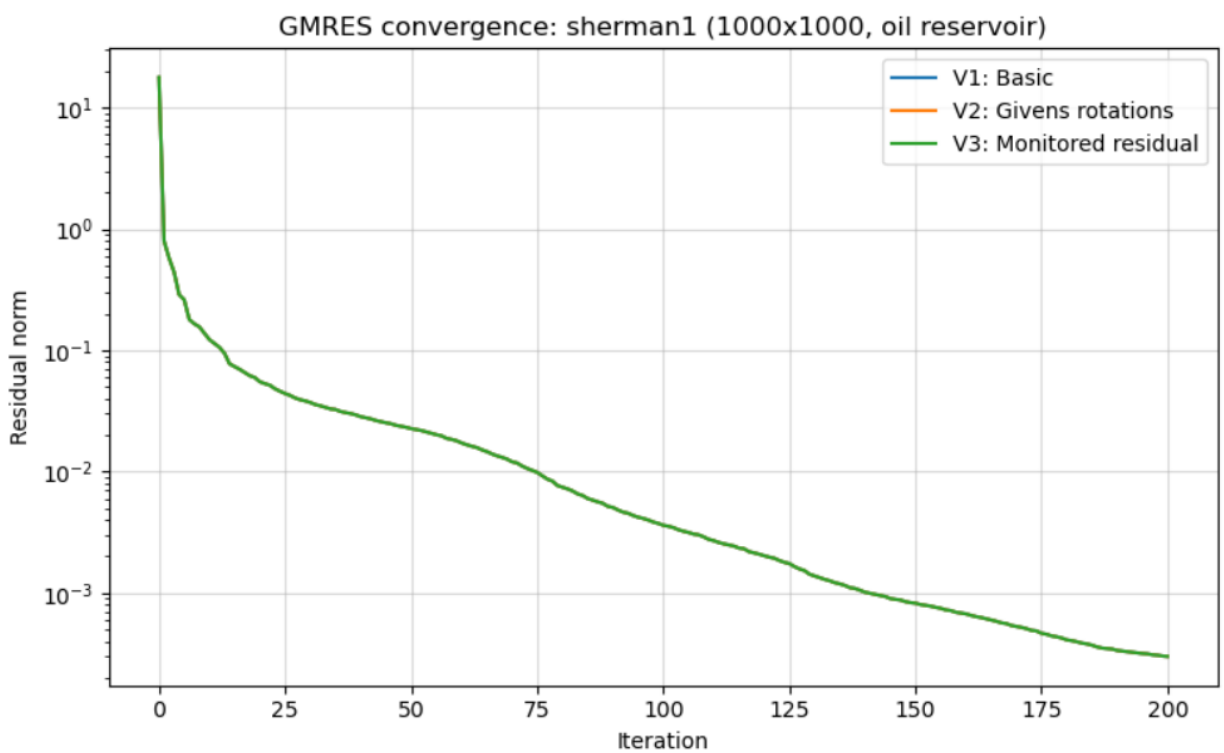

Figure 2: Convergence plot for sherman1 (1000×1000)

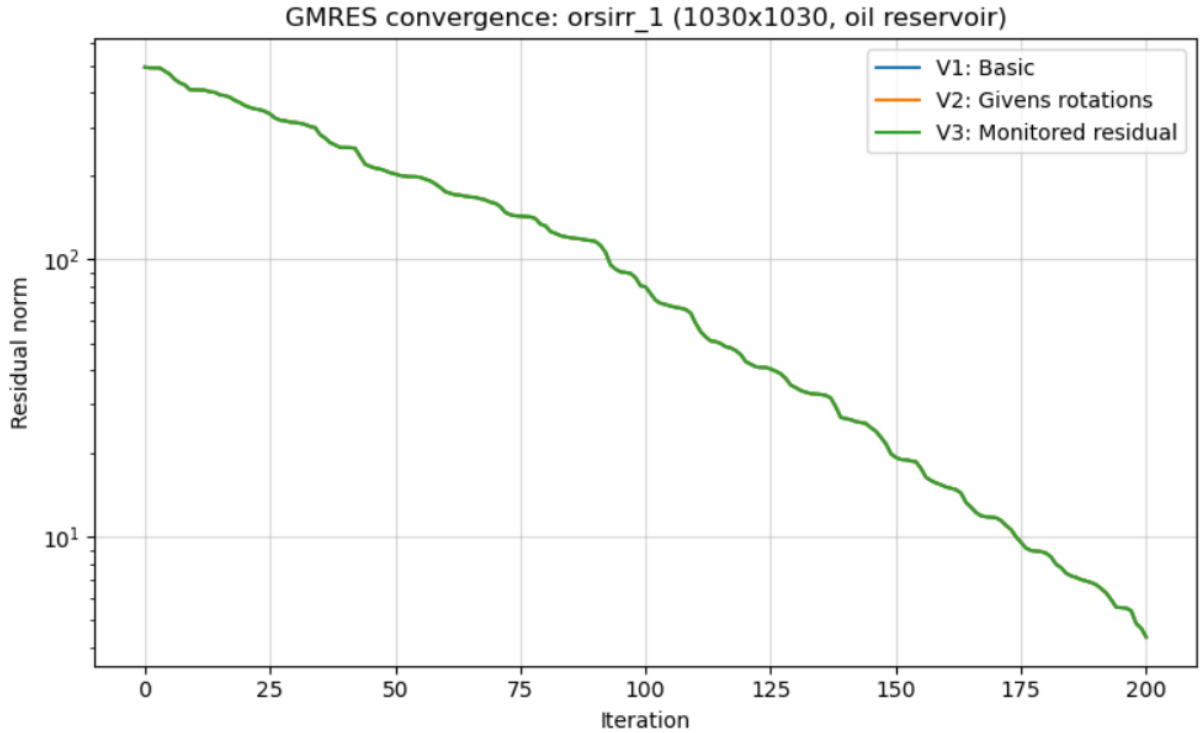

Figure 3: Convergence plot for orsirr_1 (1030×1030)

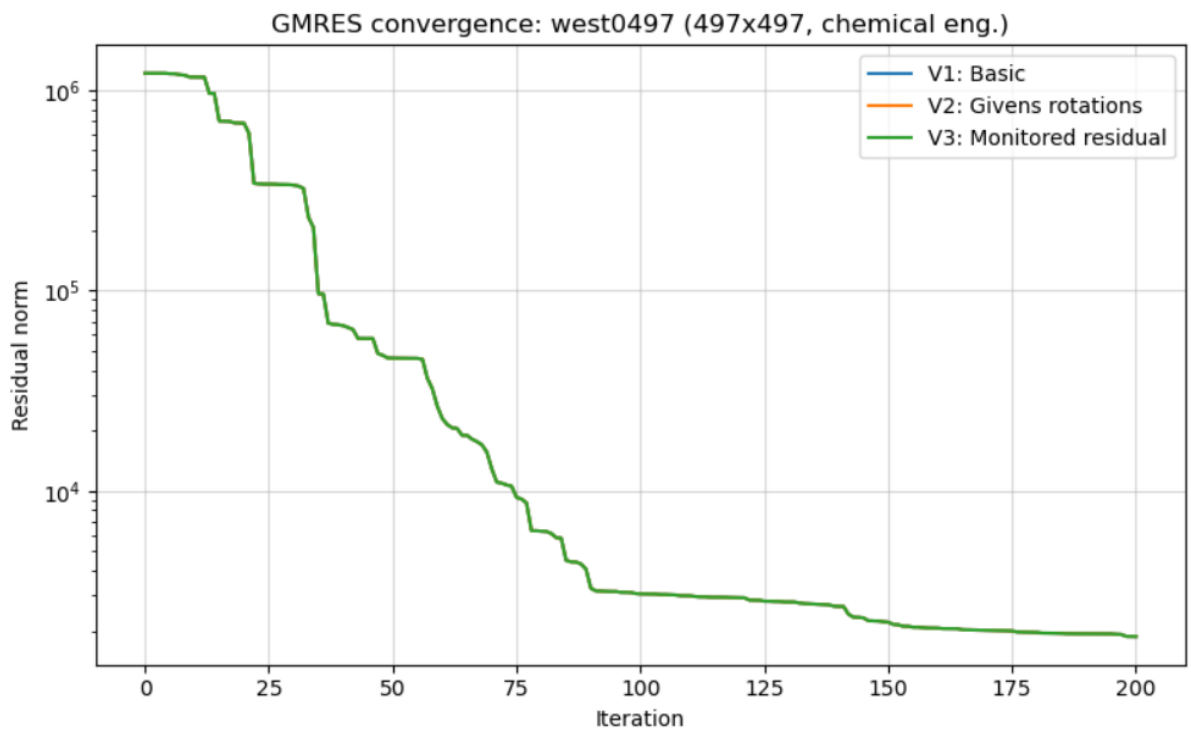

Figure 4: Convergence plot for west0497 (497×497)

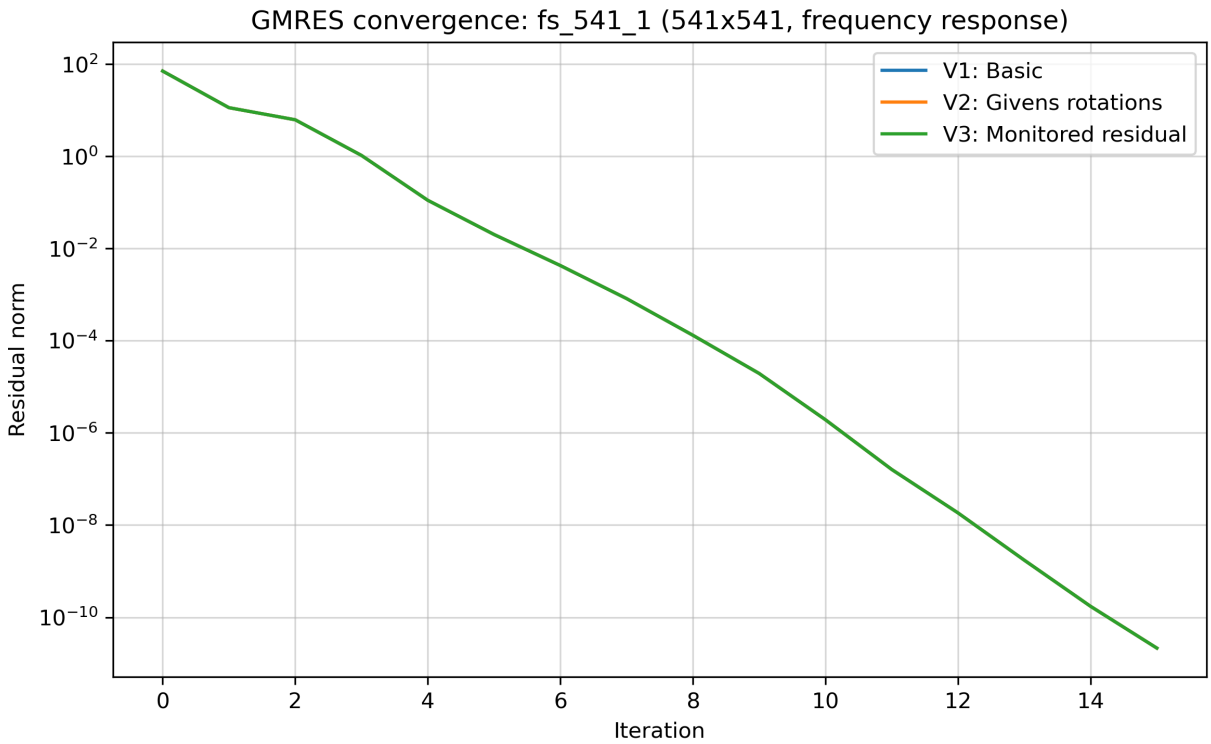

Figure 5: Convergence plot for fs_541_1 (541×541)